MAPSIGNAL 5G NETWORK MONITOR

USER GUIDE

INTRODUCTION

ABOUT MAPSIGNAL

MapSignal is a powerful yet easy-to-use mobile network monitoring tool designed to help users understand how their device connects to cellular networks. It provides real-time insights into signal strength, network technologies (from 2G to 5G), and nearby cell towers, all through an intuitive interface.
With its interactive map and advanced analytics, MapSignal allows users to visualize coverage, analyze performance trends over time, and compare network conditions across different locations. The app also includes a built-in speed test to measure download and upload performance, with results displayed through charts and historical data.

MapSignal enables users to record signal strength, quality, and performance metrics over time, making it a valuable tool for detailed analysis, troubleshooting, and drive testing scenarios. Unlike many other similar tools, **MapSignal does not require root** access, ensuring a safer and more secure user experience. It is designed with privacy in mind: the app does not collect, store, or share any personal user data. All recorded information remains exclusively on the user’s device and under their full control.
Designed for both professionals and enthusiasts, MapSignal also supports advanced use cases such as network data import and export of detailed metrics for further analysis.

PURPOSE OF THIS MANUAL

This manual provides a comprehensive guide to using MapSignal, explaining its features, functionalities, and core concepts. It is designed to help users understand how to effectively monitor mobile network performance, analyze signal quality, and interpret key network metrics.
The document covers both basic and advanced use cases, from initial setup and first use to detailed analysis and data handling. It aims to support a wide range of users, including casual users, enthusiasts, and professionals, by offering clear instructions and practical examples.

By following this manual, users will be able to fully leverage MapSignal’s capabilities to collect, visualize, and analyze mobile network data in a simple, secure, and efficient way.

GETTING STARTED

This section provides all the information required to start using MapSignal, from installation to the first interaction with the application.

INSTALLATION AND REQUIREMENTS

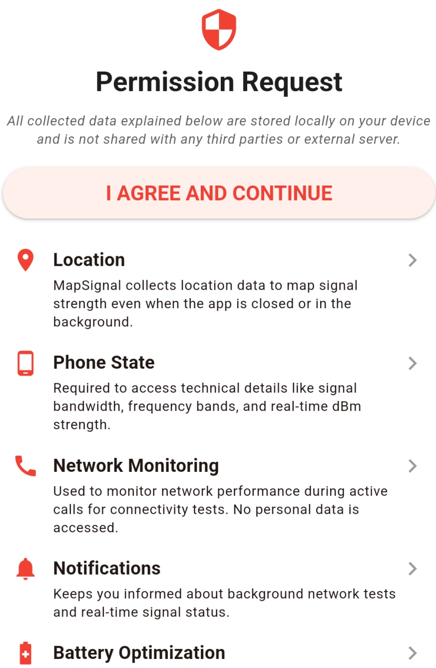
MapSignal can be installed directly from the Google Play Store. Search for "MapSignal 5G Network Monitor" and follow the standard installation process, or use the following URL:

https://play.google.com/store/apps/details?id=com.kijuzdev.mapsignal&hl=en

MapSignal supports Dual Sim devices and requires Android 11 or. later.

APP PERMISSIONS

You need to grant certain permissions for MapSignal to function properly.



MapSignal Permission Page

No user data is collected, stored, or shared. All data remains exclusively on the user's device and is fully under their control.

Below is the list of permissions required for MapSignal to operate correctly and to collect signal level data, perform speed tests, and plot measurements on the map. These permissions also enable specific features such as initiating phone calls, limited exclusively to numbers defined by the user (no automatic call answering is supported).

<table>
<tr><td>Location
• approximate location (network-based)
• precise location (GPS and network-based)</td><td>Device ID & call information
• read phone status and identity</td></tr>
<tr><td>Phone
• directly call phone numbers
• read phone status and identity</td><td rowspan="2">Other
• read precise phone states
• view network connections
• full network access
• run at startup
• control vibration
• prevent device from sleeping
• Google Play license check</td></tr>
<tr><td>Wi-Fi connection information
• view Wi-Fi connections</td></tr>
</table>

List of App Permissions

Users may choose to deny certain permissions; however, without them, the accuracy and reliability of MapSignal cannot be guaranteed.

LOCATION & WIFI

Location permission is required to access radio signal levels and cell identification.
It is also essential for recording user equipment location during test execution, mapping signal strength and quality, and correlating performance issues and network behavior.
Wi-Fi permissions are needed to accurately display the network used for the executed tests

PHONE

Phone access permission is required to access cell identification information.

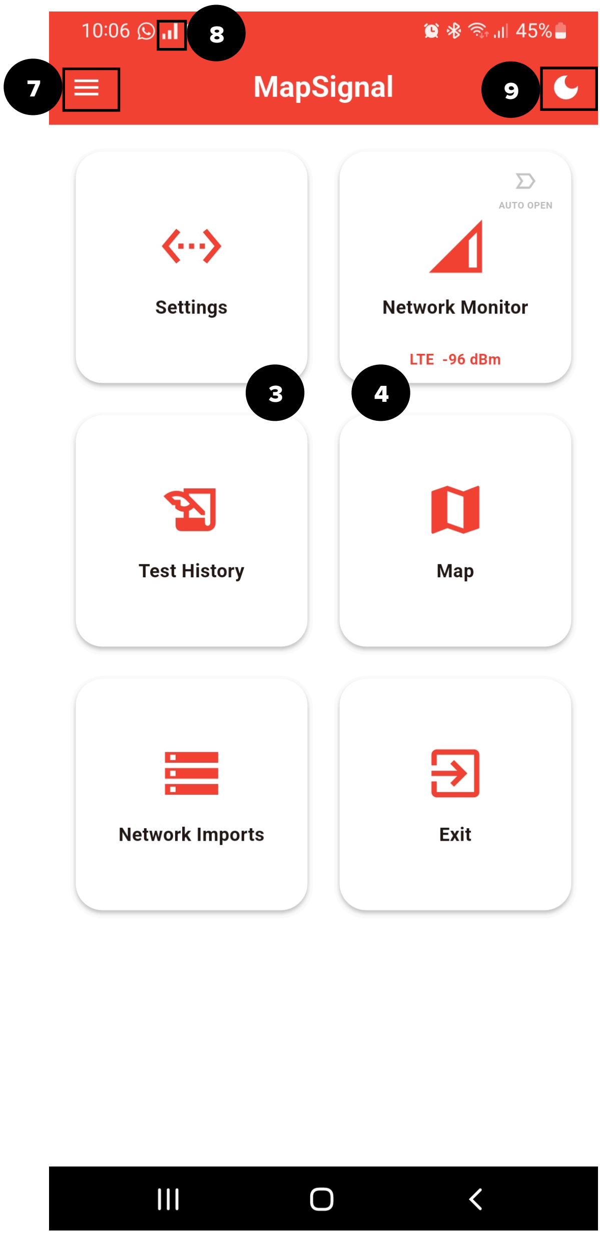
DEVICE ID & CALL INFORMATION

In addition, phone status and identity are required to initiate automatic test calls and to determine the start and end of the test call.
Please note that, for security reasons, MapSignal cannot automatically answer incoming calls.
MapSignal uses the Device ID to help you identify your device model and to organize tests on Google Drive if you use the same account across multiple devices.

OTHER PERMISSIONS

Other minor permissions are required to ensure that the app functions correctly and displays all necessary network information.

HOME PAGE OVERVIEW



Settings
This section contains the main application settings, such as 1
language preferences, server configurations, and more.

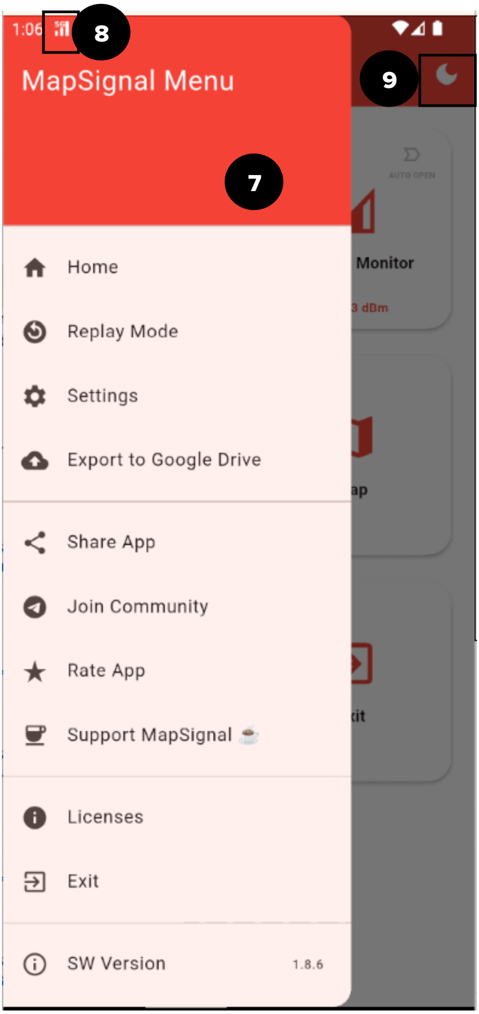
Network Monitor:
Monitor real-time network status and 2
perform signal or data performance tests.

Test History
Review your past tests, with the ability to visualize data on the 3
map or through detailed time-series charts.

Map
Visualize network data, your current location, 4
and cell towers directly on the map.

Network Imports
Upload your cell 5
tower database files in CLF or CSV format.

EXIT
Close the 6
application.



App Drawer/Menu

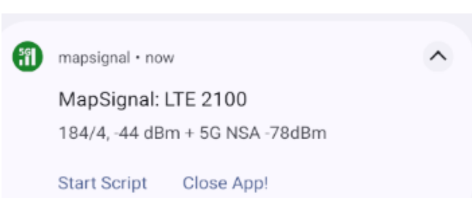
From the App Drawer you can access some quick links and other advanced modes (Replay Mode, Google Drive) 7

Notification Icon

In the App Bar you can find the notification icon. It dynamically change when a real 5G signal is available. 8

Dark/Light Mode

With this button you can quickly change the App color mode from light to dark mode. 9

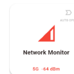


MapSignal Notification

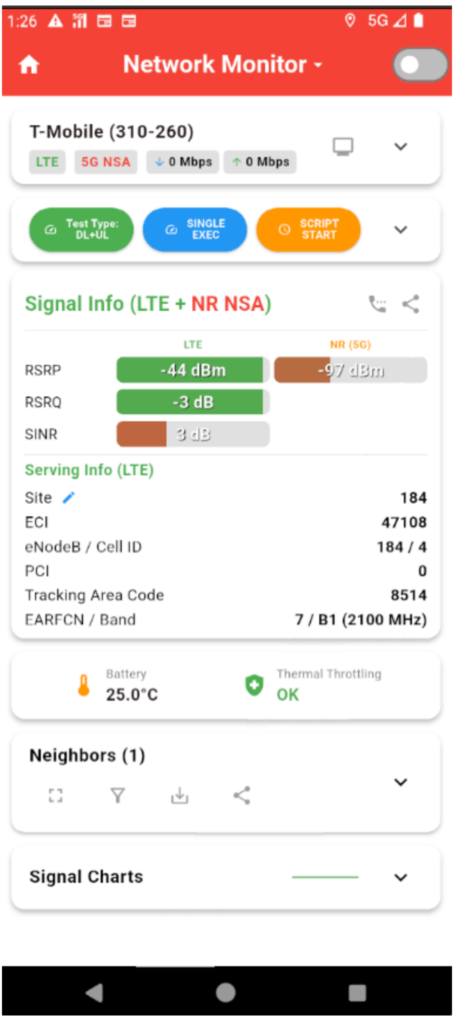
On the lockscreen you can see the MapSignal Notification, a very useful quick view of Cell id , Signal Level and System Band. Here you can find also the Test Status

QUICK START

CHECK SIGNAL LEVEL AND NETWORK INFO

From the home page, select the **Network Monitor** icon. 

The Network Monitor is organized into several sections, some of which can be expanded by clicking the small arrow on the right.



Connection Status
Operator and Connection Info + Drivetest mode

Test Section. Press <u>Test Type</u> to choose the type of test to execute.
Click <u>Single Exec</u> to start a single test or click <u>Script Start</u> to execute a series of tests (3 repetitions by default)

Signal and Serving Info Section
This Section shows you all the information about the cell your Device is currently connected to, and you can monitor the signal level and quality. When a test is ongoing all of this information are recorded and saved into an internal database and can be replayed or exported. **Note:** The Site, ECI, and eNodeB fields are highlighted in **red** if the cell is not present in your currently imported archive

Neighbors Section
Expand this card to see all neighboring cells measured by the device.

Charts Section
Expand this card and choose the chart you want to see : Signal Level, Quality, Throughput and Battery Charts

START TEST WITH DATA RECORDING

Here a simple istruction to execute a test

Expand the **Test Card**, enter a **Test Name**, and select the **Test Type**. Set the number of repetitions, test duration, and choose a **Start Time**. Finally, click **Script Start** to begin the test sequence.

Test Duration refers to each individual phase of the test. For example, if you select **DL+UL** with a **7-second** duration, the system will perform a 7-second download followed by a 7-second upload. After a customizable default delay, the next cycle will begin.

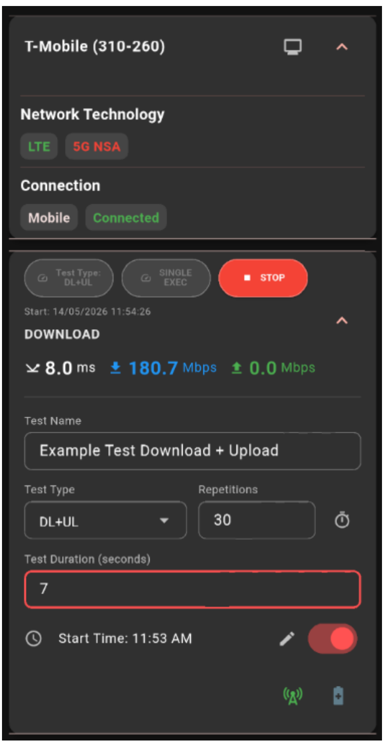
The **clock icon** next to the repetition count allows you to automatically calculate and insert the number of cycles needed to reach a specific total duration.

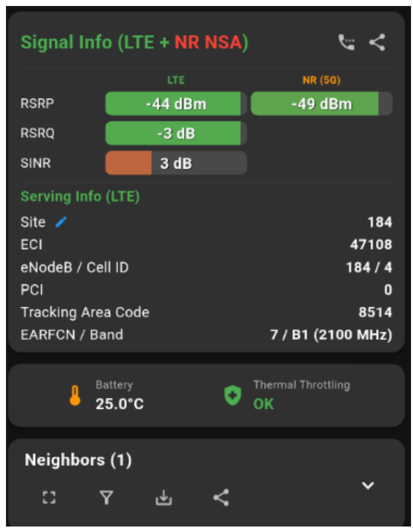
Users can schedule the test execution for a later time by using the toggle switch next to **Start Time**.

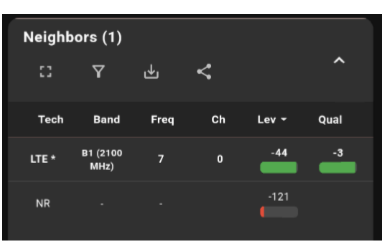
To improve data accuracy while the screen is off, we recommend enabling **'Improve NR Info Collection'**. This feature briefly wakes the screen at each test cycle, forcing Android to retrieve real-time 5G NSA information directly from the modem instead of relying on outdated or cached data.

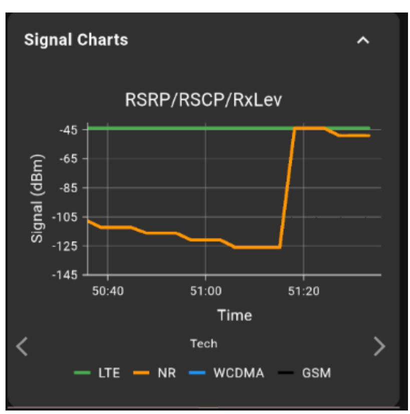
If you select **'Single Exec'** instead of 'Script Start', the system will perform only one test sequence. Just like with full scripts, the results are automatically saved to the database for later analysis.

During script execution, you can monitor signal levels, neighboring cells, and real-time signal and speed charts. All of this information is recorded during the test and saved to the database for later review and replay.
Below are some screenshots of the Network Monitor with expanded sections.





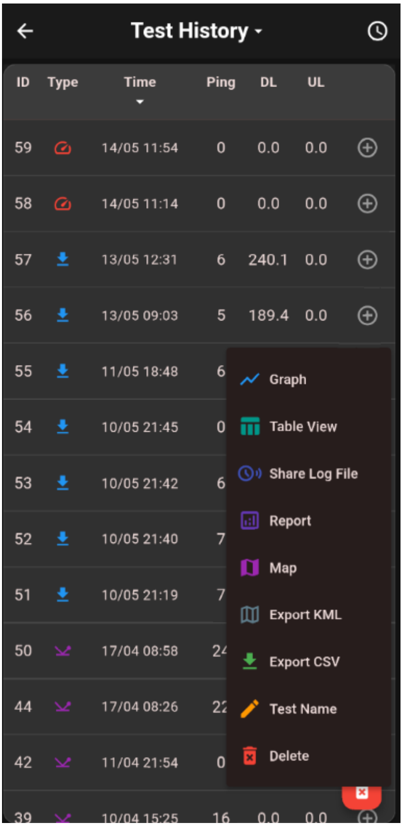




REVIEW AND REPLAY RECORDED TEST

From the home page, select the **Test History** icon.

From the **Test History** page, click on a specific test to access the following options:



1. **Graph:** Replay the log to visualize signal trends and other network-related data.

2. **Table View:** View all recorded information in a detailed tabular format.

3. **Share Log Files:** Export log files to replay the test on another device with MapSignal installed.

4. **Report:** Generate and save or share a concise summary report of the test.

5. **Map:** Replay the test on a map and plot parameters such as signal level, throughput, and more.

6. **Export KML:** Export the data to KML format to visualize signal levels on **Google Earth**.

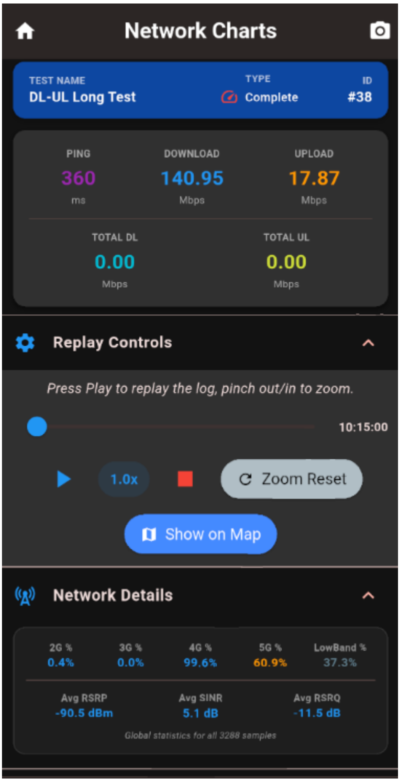
7. **Export CSV:** Export the test data in a tabular **CSV** format.

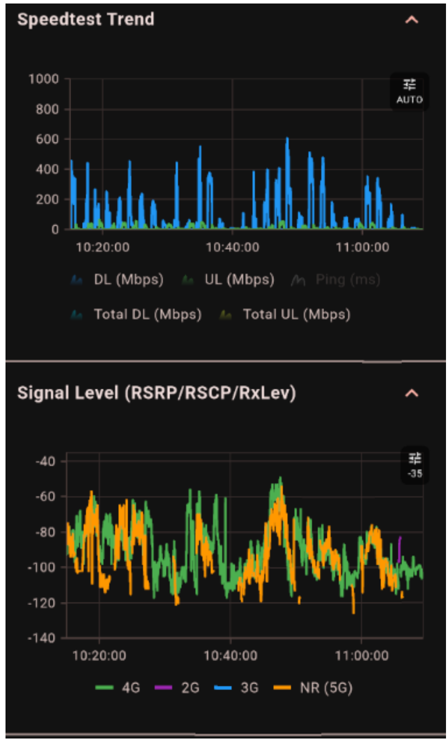
8. **Test Name:** View or edit the name of the recorded test.

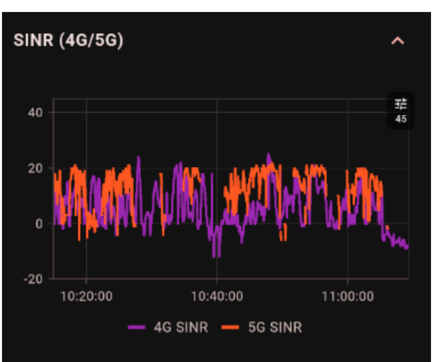
9. **Delete:** Permanently remove the recorded test from the database.

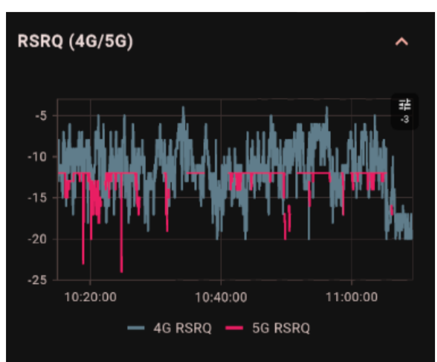
NETWORK CHARTS

This section allows you to **replay test sessions** and monitor all network metrics through an **interactive graph view**.









MAP

In the Map section, you can:

Track Test Locations: View the exact geographical points where tests were conducted.

Access Network Details: View comprehensive network information for each point.

Replay Sessions: Replay recorded tests to analyze performance over time.

Visualize Metrics: Plot signal levels, throughput, and frequency bands statistics directly on the map.

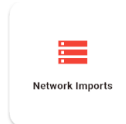
Plot Base Stations: Display Base Station locations, names, site codes, other labels and a “serving line” from your phone to serving Base Station

Show Heatmaps: Signal level and throughput heatmaps based on all tests executed by the user.

Launch Tests: Start a new script or a single test directly from the map interface (from live view only).

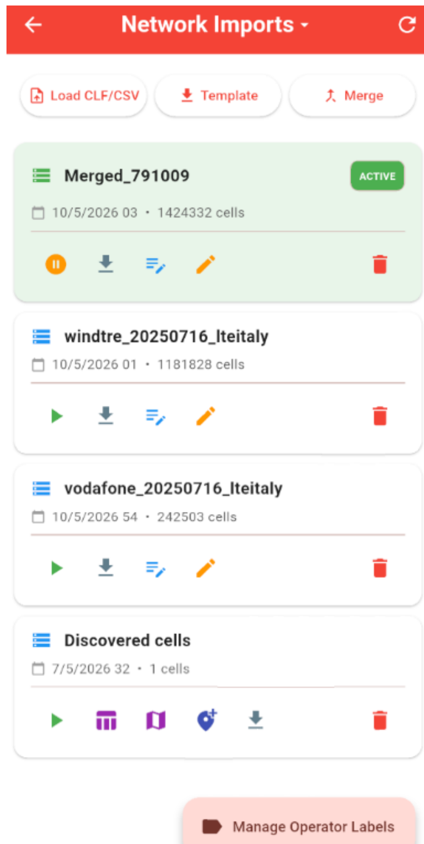
NETWORK IMPORTS

From the home page, select the **Network Imports** icon.



In this section, you can load and activate cell tower databases to visualize them on the map.

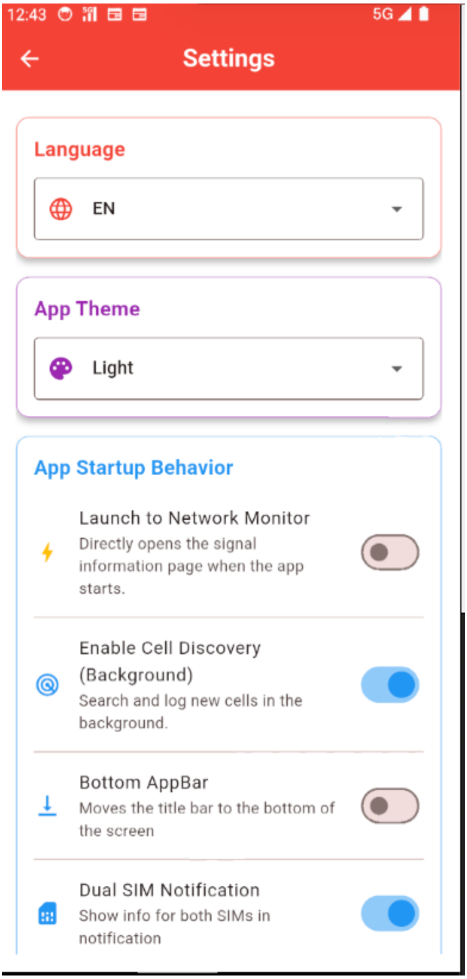
Specifically, you can:



- **Import Files:** Load CLF or CSV files in various formats.
- **Download Templates:** Get template files to create your own cell database.
- **Manage Archives:** Enable or disable loaded archives.
- **Merge Data:** Combine multiple archives containing different operator tower information.
- **Organize:** Rename or delete existing archives.
- **Edit:** Modify the content of an archive.
- **Customize:** Define custom Operator Names to use it in the Map , Notification and Monitor section

SETTINGS

This page allows you to configure advanced settings and other useful features.



Language: MapSignal automatically detects your "locale," but you can manually force a different language here.

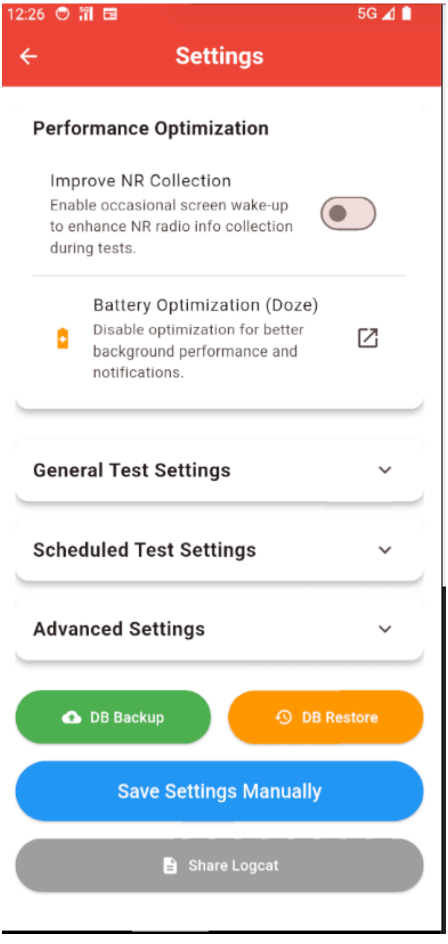
App Theme: Set the interface to Light, Dark, or System Default to match your Android theme settings.

App Startup Behavior:

•*Launch to Network Monitor*: Enable this toggle to open the app directly in the Network Monitor section upon startup.

•*Enable Cell Discovery*: This feature allows MapSignal to intelligently discover cells not present in your CLF file.
A "Discovered Cells" archive will be created in the **Network Imports** section, where you can promote new cells to your database with a single click. Discovered cells are also highlighted in the Network Monitor's "Serving Card," with an option to quickly add them to your existing archive.

- *Bottom App Bar*: Move the App Bar from the top to the bottom of the screen.
- *Dual SIM Notification:* Enable this to show information for a second SIM (if present) in the Android notification shade.



Performance Optimization

•*Improve NR Collection:* When enabled, MapSignal will occasionally wake up the screen during active script tests. This forces Android and the modem to provide real-time data for 5G NSA (Signal, connection info, etc.).

•*Battery Optimization:* Please disable Battery Optimization for MapSignal to ensure Android does not terminate the app during background operations.

General Test Settings In this section, you can define custom URLs for Upload, Download, and Ping tests, set the number of parallel downloads, configure the phone number for Voice Call tests, and more.

Scheduled Test Settings Set the number of repetitions for scripts and the interval (in seconds) between sequences.

Advanced Settings

5G Cell Decoding: Unlike LTE, 5G cell decoding is not fixed. Operators can vary the number of bits used for gNodeB/Cell ID translation from the NCI. You can manually define this parameter or allow MapSignal to detect it automatically.

DB Backup/Restore: Use this to back up your test history. This is essential when reinstalling MapSignal or migrating to a new phone.

Share Logcat: Generate an advanced system log to share with the developer for bug reporting and troubleshooting.

COMMUNITY & SUPPORT

Support MapSignal :

MapSignal is an independent project built for network analysis and optimization.
If you find it useful in your work or testing activities, consider supporting its development with a coffee ☕
Your support helps maintain the app, improve performance, and deliver new features.
This is a voluntary donation.
Thank you!

https://www.paypal.com/ncp/payment/5Q23FACCT7HJU

Telegram Community:

https://t.me/mapsignal_community

YouTube Channel:

https://www.youtube.com/@gb_dev-w7f